INCH-POUND

MIL-I-52701B(ME)
12 March 1991
SUPERSEDING
MIL-I-52701A
19 September 1978

MILITARY SPECIFICATION

INDICATOR SOLUTION, pH DETERMINATION

(RANGE 3.0 TO 11.0)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an indicator solution for the determination of pH values of water in the range 3.0 to 11.0.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-C-2020 - Chemicals, Liquid, Dry, and Paste: Packaging of.

Beneficial comments, (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6810

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STANDARDS

FEDERAL

FED-STD-313 - Material Safety Data Sheets, Preparation and the Submission of.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 - Marking for Shipment and Storage.

MIL-STD-1190 - Minimum Guidelines for Level C Preservation, Packing and Marking.

MIL-STD-1472 - Human Engineering Design Criteria for Military Systems, Equipment and Facilities.

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

ANSI Z129.1 - Precautionary Labeling of Hazardous Industrial Chemicals.

(Application for copies should be addressed to American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018).

AMERICAN PUBLIC HEALTH ASSOCIATION (APHA)

Standard Methods for the Examination of Water and Wastewater, 16th Edition, 1985.

(Application for copies should be addressed to the American Public Health Association, 1015 Fifteenth Street, N. W., Washington, DC 20005.)

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NATIONAL BUREAU OF STANDARDS (NBS)

Research Paper 2619 - pH Values of the Clark and Lubs Buffer Solutions at 25 °C.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402).

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The indicator solution shall produce a color for specific pH values of water that visually match color standards of a master color disk. The solution shall be suitable for use in determining pH values of water from 3.0 to 11.0 in increments of 1 pH when tested in accordance with 4.3.3.

3.2 Standard sample. A standard sample of indicator solution will be supplied by the Government on request by a potential bidder. The standard sample will be furnished only to show the potential bidder the color (obtained when 0.50 mL of standard sample is added to 15 mL of each buffer solution prepared in accordance with 4.3.3.2) that the indicator solution he proposes to furnish must produce (see 6.4).

3.3 Material. Material shall be as specified herein. Material not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Suspended matter. Total suspended solids shall not exceed 10 mg/L. When specified (see 6.2), the contractor may certify that the indicator solution has been filtered through Whatman grade 934AH and 984H, Gelman type A/E, Millipore Type AP40, or equivalent glass fiber filters prior to packaging.

3.4 Material Safety Data Sheets. A Material Safety Data Sheet shall be prepared in accordance with the requirements of FED-STD-313 (see 6.2.1.h).

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3.7 Government-loaned property. Unless otherwise specified (see 6.2), the following property in the quantities indicated will be loaned by the Government (see 6.3):

Item No.	Description	Quantity
1.	Disk, master color standard, medium values, pH 3.0 to 11.0	1
2.	Disk, master color standard, low values, pH 3.0 to 11.0	1
3.	Disk, master color standard, high values, pH 3.0 to 11.0	1
4.	Disk evaluator.	1

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for ensuring that components and materials are manufactured, examined, and tested in accordance with referenced specifications and standards as applicable.

4.2. Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Quality conformance inspection (see 4.3).
b. Inspection of packaging (see 4.4).

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4.3 Quality conformance inspection.

4.3.1 Sampling. Sampling for examination and test shall be in accordance with MIL-STD-105, tables I and IIa. A lot shall be accepted if 0 defects are found and rejected if 1 or more defects are found.

4.3.2 Examinations. Samples selected in accordance with 4.3.1 shall be examined for the following defect:

102. Material Safety Data Sheet not prepared in accordance with FED-STD-313 (see 3.4).

4.3.3 Tests.

4.3.3.1 Suspended matter. Unless otherwise specified, samples selected in accordance with 4.3.1 shall be tested for suspended matter or foreign material in the solution in accordance with section 209C, (Total Suspended Solids Dried at 103 - 105 °C) per APHA Standard Methods for the Examination of Water and Wastewater. Nonconformance to 3.3.1 shall constitute failure of this test.

4.3.3.2 Color. Samples selected, tested and approved per 4.3.2 or, when certification is specified, samples selected in accordance with 4.3.1, shall be tested for accuracy of color intensity produced. Buffer solutions of pH 3.0 to 11.0 inclusive in increments of 1 pH shall be prepared in accordance with NBS Research Paper 2619. Add 0.50 mL of indicator solution to 15 mL of each buffer solution and compare each color produced with Government-loaned low, medium, and high master color standard disks. Failure of the color produced by the indicator solution to match within the range of color of the low and high value master color standard disks shall constitute failure of this test.

4.4 Inspection of packaging.

4.4.1 Unit of product. For the purpose of inspection, a complete package of indicator solution prepared for shipment shall be a unit of product.

4.4.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105, tables I and IIa. A lot shall be accepted if 0 defects are found and rejected if 1 or more defects are found.

4.4.3 Examination. Samples selected in accordance with 4.4.2 shall be examined for the following defects:

101. Preservation not as specified for level A or C (see 5.1.1 and 5.1.2)
102. Packing not as specified for level A, B, or C (see 5.2.1 or 5.2.2)
103. Marking missing, illegible, incomplete, or not as specified (see 5.3).

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5. PACKAGING

5.1 Preservation. Preservation shall be level A or C, as specified (see 6.2).

5.1.1 Level A. The indicator solution, in quantities of 4 ounces or 16 ounces, as specified (see 6.2), shall be preserved in accordance with level A preservation requirements of PPP-C-2020, using type III, class 3 containers.

5.1.2 Level C. The indicator solution, in quantities of 4 ounces or 16 ounces, as specified (see 6.2), shall be preserved for level C in accordance with the commercial preservation requirements of PPP-C-2020 using plastic containers.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A or B. Packing shall be in accordance with PPP-C-2020 for the level specified.

5.2.2 Level C. Packing shall be in accordance with MIL-STD-1190.

5.3 Marking. In addition to any special or identification marking required by the contract or purchase order (see 6.2), all containers shall be marked in accordance with MIL-STD-129 and as specified herein (see 5.3.1). The markings shall be located to conform to the requirements of MIL-STD-1472, 5.5.

5.3.1 Identification and precautionary labels. Containers as specified herein (see 5.1) shall have labels, as applicable in accordance with ANSI Z129.1 and shall meet the criteria of MIL-STD-1472, 5.5.

5.4 Unitization. When specified (see 6.2), unitization of shipping containers shall be as follows:

5.4.1 Levels A and B containers. Unitization shall be in accordance with the requirements of PPP-C-2020.

5.4.2 Level C containers. Unitization shall be in accordance with MIL-STD-1190.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The indicator solution covered by this specification is intended for use in determining the pH of water by comparison with color disks conforming to MIL-D-537, type I, class B.

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. When Government-loaned property is not required (see 3.7).
c. When certification is accepted (see 4.3.2.1).
d. Level of preservation and packing required (see 5.1 and 5.2).
e. Whether unit container quantity shall be 4 ounces or 16 ounces (see 5.1.1 and 5.1.2).
f. Any special markings (see 5.3).
g. When unitization is required (see 5.4).
h. Preparation of Material Safety Data Sheet (see 3.6).

6.3 Government-loaned property. The contracting officer should arrange to loan property specified in 3.7, when applicable (see 6.2).

6.4 Standard sample. The contracting officer should arrange to furnish potential bidders a 16-ounce bottle of indicator solution, pH wide range, NSN 6810-00-087-2340.

6.5 Material Safety Data Sheets. Contracting officers should identify those activities requiring copies of completed Material Safety Data Sheets prepared in accordance with FED-STD-313. The pertinent Government mailing addressed for submission of data are listed in appendix B of FED-STD-313.

6.6 Subject term (key word) listing.

Chemical solution

6.7 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes which respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - ME

Review activities:
Army - MD, AR
DLA - GS

Preparing activity:
Army - ME

Project 6810-AB38

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-I-52701B (ME)	12 March 1991

3. DOCUMENT TITLE
Indicator Solution, pH Determination (Range 3.0 to 11.0)

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
	(703) 664-5717	354-5717
c. ADDRESS (Include Zip Code) US Army Belvoir RDE Center ATTN: STRBE-TSE Ft. Belvoir, VA 22060-5606	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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